

Term Project Proposal: Text- and Night-Light-Guided Satellite Embeddings for Regional Economic Estimation

Sep 29, 2026 · @Hemant

1. Summary

This project builds and tests a pipeline that estimates district-level GDP per capita from satellite data, by combining the ideas of two papers: UrbanCLIP (Yan et al., 2024) and Estimating Economic Activity from Satellite Embeddings (Yue, Zhao and Hu, 2026).

From UrbanCLIP we take the idea of adding **text**: a vision-language model writes a description of each satellite tile, and a model learns to link images with their descriptions. From the embedding paper we take two ideas: using Google's ready-made **satellite embeddings** instead of raw images, and using **nighttime lights** as a cheap stand-in label that steers those embeddings toward economic meaning.

The pipeline climbs a ladder of supervision. Stage A uses no labels (image-text matching), Stage B uses a cheap proxy label (nighttime lights), and Stage C uses the scarce real label (official district GDP). Stage D tests the result honestly with spatial cross-validation, ablations and an error analysis.

The main deliverables are a working, reproducible pipeline, a comparison table showing how much each ingredient (text, night-light steering, POI and roads) adds, a 1 km map of estimated economic activity, and a short error analysis showing where the model works and where it fails.

2. Background and motivation

Official economic statistics are often missing, late or unreliable, especially below the national level and in countries with weak statistical systems. Satellites see the whole planet every year at the same quality, so they offer an independent way to estimate where economic activity is and how it changes (see Concept 3.1).

2.1 What the two source papers did

UrbanCLIP (Yan et al., WWW 2024) asked whether text can improve urban region profiling. It used LLaMA-Adapter V2 to caption about 17,500 satellite tiles (about 1 km each) in Beijing, Shanghai, Guangzhou and Shenzhen, cleaned the captions, and trained an image-text model with a contrastive loss plus a captioning loss. A frozen image encoder then fed a small predictor for carbon emissions, population and GDP. It beat the best baseline by about 6.1% in R^2 on average.

Estimating Economic Activity from Satellite Embeddings (Yue, Zhao and Hu, Applied Sciences 2026) used Google's precomputed 64-number satellite embeddings (GSED, from AlphaEarth Foundations). A small network learned to predict nighttime lights from the embeddings, and its 32-number middle layer became an "income-aware" embedding. These were summed per country to predict GDP. The embedding model beat nighttime lights for GDP levels (MSE 0.504 vs 0.602), a blend of the two did best (0.459), and nighttime lights stayed better for growth.

2.2 Gaps this project addresses

- **UrbanCLIP's GDP gain was tiny** (+0.52% R^2 on average), and it split tiles randomly, so neighbouring near-identical tiles could fall in both training and test sets (leakage, Concept 3.9).
- **UrbanCLIP's caption cleaning relied on manual checking** by students, which does not scale.
- **UrbanCLIP's GDP label was itself built from nighttime lights**, so part of the "ground truth" is a satellite product (Concept 3.10).
- **The embedding paper used no text and no POI or road data**, and worked only at country level, where there are few training examples.
- **Neither paper tested the two ideas together.** We do not know whether text alignment and night-light steering help each other, overlap, or conflict.

This project fills the last gap directly and reduces the others, at a scale that is realistic for one term.

3. Key concepts and theory

Each concept below is explained in plain language, followed by where it is used in this proposal. Section numbers point to the rest of the document.

3.1 Satellites as a proxy for the economy

A *proxy* is something we can measure that moves together with something we cannot measure directly. Richer, busier places tend to have more buildings, roads, lights and industry, and these are visible from space. Henderson, Storeygard and Weil (2012) made this idea standard in economics using nighttime lights.

Used in: §2 (motivation), §7.4 (Stage B), §7.6 (hybrid estimator).

3.2 Nighttime lights (NTL) and their limits

NTL satellites (here VIIRS) measure how bright each place is at night. Brightness tracks electricity use, infrastructure and activity, so it correlates with income. Two known problems: it **saturates** (very bright cities all look the same, very poor rural areas all look dark) and it misses economic activity that does not produce light, such as services or farming.

Used in: §7.4 (NTL as a stand-in label), §7.6 (NTL-only benchmark), §8.4 (why errors grow in rich areas).

3.3 Embeddings

An *embedding* is a list of numbers that summarises something complex, such as an image or a sentence. It is built so that similar things get similar lists. We can measure similarity with *cosine similarity*, which checks whether two lists point in the same direction.

Used in: §7.1 (GSED embeddings per grid cell), §7.3 (image and text embeddings), §7.7 (finding similar places).

3.4 Foundation models and Google's satellite embeddings (GSED)

A *foundation model* is a large model trained once on huge amounts of general data, then reused for many tasks. Google's AlphaEarth Foundations model compresses a full year of Sentinel-1 radar and Sentinel-2 optical images into 64 numbers per 10 m pixel. The result is released as the Satellite Embedding dataset (GSED), covering 2017 onward. We do not need to process raw images ourselves.

Used in: §6 (data), §7.1 (main image input), §7.3 (image side of Stage A).

3.5 Vision-language models (VLMs) and hallucination

A *VLM* is a language model that can look at an image and write about it (for example Qwen2.5-VL, LLaVA, Florence-2). UrbanCLIP used LLaMA-Adapter V2. VLMs sometimes *hallucinate*: they confidently describe things that are not there, such as calling a highway a river. The quality of the text therefore needs checking.

Used in: §7.2 (caption generation and cleaning).

3.6 Contrastive learning

Contrastive learning trains a model with matching and non-matching pairs. The model is rewarded when an image and its own caption end up close together, and when an image and a random other caption end up far apart. CLIP (Radford et al., 2021) made this popular; UrbanCLIP uses it for satellite images.

Used in: §7.3 (Stage A).

3.7 Transfer learning, freezing and fine-tuning

Transfer learning means reusing what a model learned on one task to help another. A *frozen* model is used as-is: we read its outputs but never change it. *Fine-tuning* means continuing to train an existing model on new data. A *projection network* is a few small layers that convert one list of numbers into another.

Jean et al. (2016) showed the classic version for poverty mapping: train a model to predict nighttime lights from daytime images, then reuse its internal features to predict poverty. The embedding paper applies the same trick to GSED.

Used in: §7.3 (frozen encoders plus trained projections), §7.4 (keeping the middle layer of the NTL predictor).

3.8 Learning from aggregate labels (sum-pooling)

Often the answer (GDP) is known only for a whole district, not for each 1 km cell. *Sum-pooling* handles this: the model makes a small output for every cell, adds them up for the district, and compares the total with the district's official figure. It learns cell-level patterns from district-level answers, and as a bonus can produce a fine map.

$$\hat{y}_{\text{district}} = h\left(\sum_{i \in \text{district}} p(z_i)\right)$$

Here z is a cell's embedding, p is the per-cell network and h is the final network.

Used in: §7.5 (Stage C).

3.9 Spatial autocorrelation, leakage and spatial cross-validation

Spatial autocorrelation is Tobler's first law: nearby places are more alike than distant ones. If the data are split randomly, a test cell usually has a near-identical neighbour in the training set, so the model can "cheat". This is *data leakage*. *Spatial cross-validation* fixes it by splitting whole geographic chunks (here, groups of districts) so test areas are truly unseen.

Used in: §8.1 (how data are split), §2.2 (gap in UrbanCLIP).

3.10 Measurement error and circular labels

Official GDP is not the true GDP; it is the true value plus a statistical error. Satellite estimates also have their own error. The embedding paper writes this as:

$$y^{\{\text{official}\}} = y^{\{\text{true}\}} + \varepsilon^{\{\text{stat}\}}, \quad \text{qqquad} \\ \text{residual} \approx \varepsilon^{\{\text{sat}\}} + \varepsilon^{\{\text{stat}\}}$$

So a large gap between our estimate and the official number can come from our model or from the official number. A related trap is *circular labels*: some popular "ground truth" maps (for example WorldPop population or some gridded GDP products) were partly made from nighttime lights. Using them as targets after training on NTL would mean predicting NTL from NTL.

Used in: §6 (label choice), §8.4 (error analysis).

3.11 Evaluation metrics

- **R²** (coefficient of determination): the share of variation in the target that the model explains. 1 is perfect, 0 is no better than guessing the average.
- **RMSE** (root mean squared error): typical error size, with large errors punished more.
- **MAE** (mean absolute error): average error size, treating all errors equally.

We predict log GDP per capita, so an error of 0.1 means roughly a 10% miss.

Used in: §8.2.

3.12 Ablation study

An *ablation* removes one ingredient and retrains, to see how much that ingredient mattered. Both source papers use this.

Used in: §8.3 (text, NTL steering, POI and roads, caption cleaning).

3.13 Ensembles for uncertainty

An *ensemble* trains several copies of the same model with different random starting points. Where the copies disagree, the prediction is less certain. This gives a simple uncertainty range per district.

Used in: §8.5.

3.14 Levels versus growth, and fixed effects

A *level* question asks how rich a place is; a *growth* question asks how fast it is changing. *Fixed effects* remove each region's permanent average so that only changes over time are tested. The embedding paper found embeddings strong for levels and weak for growth.

Used in: §5 (levels are the main focus), §8.6 (growth as an optional extension).

4. Objectives, research questions and hypotheses

The main objective is to test whether combining text alignment (UrbanCLIP) with night-light steering (embedding paper) estimates district GDP per capita better than either idea alone, under an honest spatial evaluation.

4.1 Specific objectives

1. Build a reproducible pipeline from free satellite embeddings, auto-generated text, map data and nighttime lights to district-level GDP estimates.
2. Replace UrbanCLIP's manual caption checking with an automatic, data-based cleaning step and report its error rate.
3. Measure how much each ingredient adds through ablations.
4. Show where the model fails and whether failures relate to income level or data quality.
5. Produce a 1 km map of estimated economic activity with short text explanations.

4.2 Research questions and hypotheses

ID	Research question	Hypothesis	Tested in
RQ1	Does night-light steering make satellite embeddings better at predicting GDP?	Yes. Steered embeddings give lower RMSE than raw GSED embeddings, as in the embedding paper.	§8.3
RQ2	Does adding text (image-text alignment) help beyond image-only embeddings?	A small gain. UrbanCLIP's GDP gain was only +0.52% R^2 , so we expect modest improvement.	§8.3
RQ3	Do POI and road data add information that images miss?	Yes, mainly for service-heavy urban districts, where images and lights are weakest.	§8.3
RQ4	Does blending our model with an NTL-only model beat both?	Yes, as in the embedding paper, with the best weight on our model around 0.5 to 0.75.	§8.3
RQ5	Where are errors largest?	Errors grow with income (services are hard to see from space) and shrink where official data are more reliable.	§8.4

A negative answer to any question is still a valid result, provided it is measured honestly.

5. Scope and study design

The project estimates **GDP per capita levels for districts in one study region, for one main year**, using 1 km grid cells as building blocks. Everything else is kept small enough to finish in one term.

5.1 Design choices

Choice	Decision	Reason
Study area	One country or group of states/provinces with at least 150 districts that have official district GDP	Stage C learns from districts, so it needs enough of them; 150+ is a practical minimum for a small network.
Spatial unit (building block)	1 km × 1 km grid cells	Matches UrbanCLIP's tile size; fine enough to see neighbourhoods.
Unit with labels	District	Official GDP is usually published per district, not per cell (Concept 3.8).
Main year	One year between 2017 and 2024 with district GDP available	GSED and VIIRS both cover 2017–2024.
Target	Log GDP per capita per district	Log scale makes errors comparable across rich and poor districts.
Question type	Levels (how rich), not growth	Embeddings are strong for levels and weak for growth (Concept 3.14). Growth is an optional extension.
Captions	A sample of about 10,000 cells, drawn only from training districts	Captions are needed only to train Stage A. At prediction time only the image side is used, so no captions are needed for the other cells.

5.2 In scope

- Stages A to D as described in §7 and §8, using Option A1 (frozen encoders plus small trained projections).
- Text from an open VLM plus template sentences built from POI, road and land-cover data.
- NTL-only benchmark and hybrid blend.

- Five-fold spatial cross-validation, ablations, error analysis and an ensemble-based uncertainty range.
- A 1 km map and a small set of explained example districts.

5.3 Out of scope (or optional if time allows)

- Fine-tuning large image or text encoders (UrbanCLIP's full model, Option A2).
- A captioning (language-modelling) loss.
- Multi-year growth estimation (optional extension, §8.6).
- Other targets such as population or carbon emissions.
- A web application like UrbanCLIP's Urban Insights.

5.4 Open question

The final study area depends on which region has reliable, downloadable district GDP. This should be confirmed with the instructor in week 1.

6. Data sources

All inputs are free. Everything is aligned to the same 1 km grid and the same year.

Layer	Role in pipeline	Source	Processing	Inspired by
Satellite embeddings (GSED)	Main image input	Google Earth Engine, dataset GOOGLE/SATELLITE_EMBEDDING/V1_ANNUAL	Average the 64 numbers of all 10 m pixels inside each 1 km cell	Embedding paper (they aggregated to 10 km; we use 1 km)
High-resolution imagery tiles	Input to the VLM for captions only	Free basemap tiles (e.g., Esri World Imagery); check the licence	One image per sampled cell, about 1 m per pixel	UrbanCLIP (Baidu Map tiles)
Nighttime lights	Stand-in label for Stage B; NTL-only benchmark	VIIRS annual composites on Google Earth Engine	Mean radiance per cell, then $\log(1 + \text{value})$	Embedding paper; Henderson et al. (2012)
Land use / land cover	Cell features; fact-checking captions; masking	ESA WorldCover (10 m)	Percent of each class per cell (built-up, cropland, water, trees...)	Embedding paper (used MODIS land cover to mask cells)
POI	Extra information on services and amenities	OpenStreetMap	Count per category per cell (schools, hospitals, banks, shops, factories)	UrbanCLIP §3.5 (suggested POI as optional extra); PG-SimCLR baseline

Layer	Role in pipeline	Source	Processing	Inspired by
Road network	Extra information on connectivity	OpenStreetMap	Road length per cell by type (major, minor)	UrbanCLIP §3.5 (road networks suggested)
District boundaries	Assign cells to districts	Official boundaries from the statistics office (fallback: GADM)	Each cell assigned to the district containing its centre	Embedding paper (GADM for countries)
District GDP	Target label (Stage C)	National or state statistics office	Convert to per capita, then log	Embedding paper (World Bank GDP per capita)
District population	Convert GDP to per capita; convert predictions back	Census or official estimates	Same year as GDP where possible	Embedding paper (World Bank population)

6.1 Caution: avoid circular labels

Our labels must not be built from nighttime lights, because Stage B already trains on NTL (Concept 3.10). Official district GDP from statistics offices is usually safe. Gridded products such as WorldPop population or NTL-based gridded GDP are **not** used as targets. UrbanCLIP's GDP label came from such a product, which is one reason its results are hard to interpret.

6.2 Caution: match years and resolutions

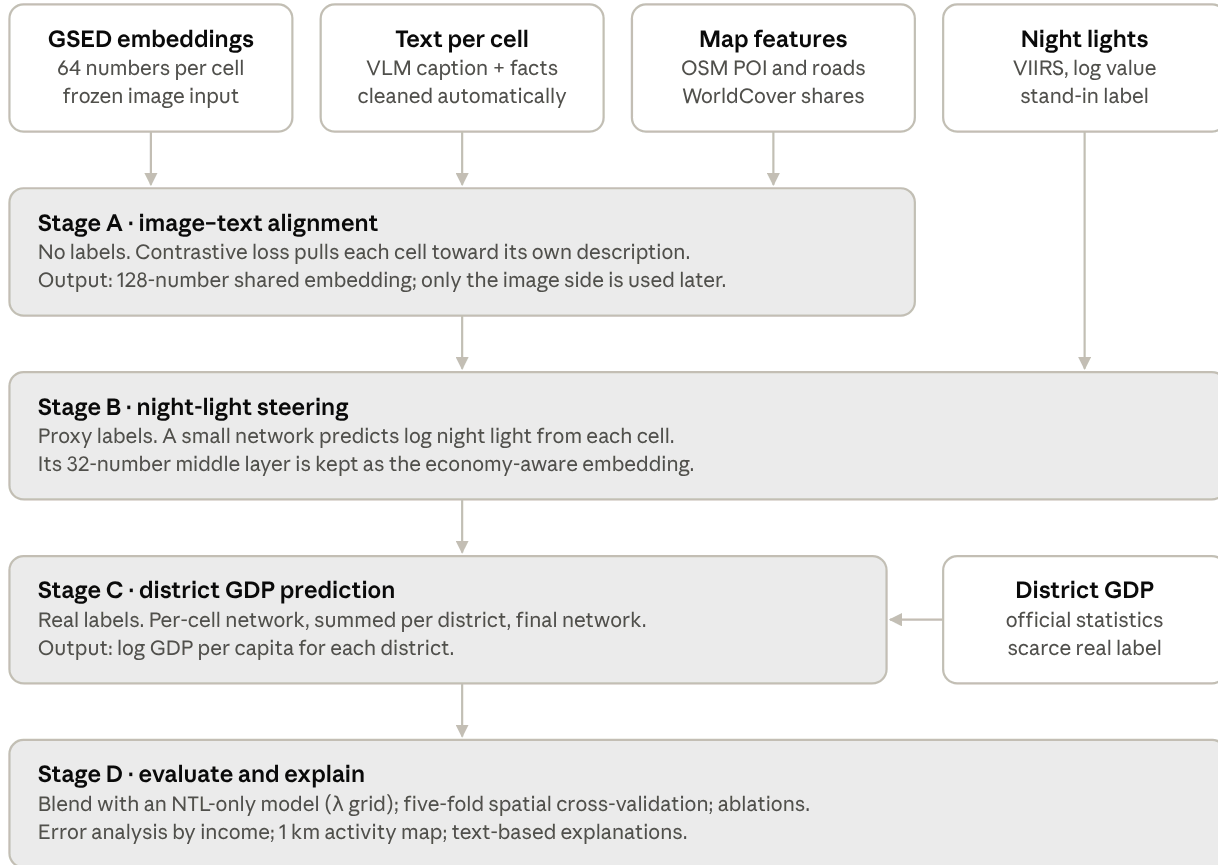
The GSED year, VIIRS year and GDP year should be the same. The high-resolution tiles may be from a slightly different year; this only affects caption quality, not the main input.

7. Methodology

The pipeline has seven steps. Steps 1–2 prepare data, Steps 3–5 are the three learning stages (no labels, proxy labels, real labels), and Steps 6–7 combine and explain the results. Each step ends with an **Inspired by** note.

Each stage uses scarcer labels: none, then night lights, then district GDP

Plain boxes are data; shaded boxes are learning stages, all retrained per fold on training districts only



pipeline · 4 inputs, 4 stages

Text and map features enter at Stage A, night lights at Stage B, and district GDP only at Stage C.

7.1 Step 1: Build the grid and the cell feature table

1. Cover the study area with a 1 km grid. Assign each cell to a district.
2. Remove cells that are almost entirely water, dense forest, snow or desert, using ESA WorldCover.
3. For each remaining cell, compute: the 64 averaged GSED numbers, log nighttime light, land-cover percentages, POI counts by category, and road length by type.
4. Save one table: one row per cell, one column per feature, plus the district ID.

Inspired by: the embedding paper (grid aggregation, land-cover masking, GADM assignment). POI and roads come from UrbanCLIP's §3.5, which suggests adding them as optional extra data.

7.2 Step 2: Generate and clean text descriptions

2a. Visual captions. For about 10,000 sampled cells, download a high-resolution tile and ask an open VLM (such as Qwen2.5-VL or Florence-2) to describe it. The prompt focuses on buildings, roads, green space, water, industry and human activity.

2b. Fact sentences. Turn each cell's numbers into template sentences, for example: "This area is 65% built-up, with 12 schools, 4 hospitals and 8 km of major roads." These come straight from data, so they cannot hallucinate.

2c. Automatic cleaning.

- Remove vague filler sentences (for example "The image offers a comprehensive view of the city") with simple text rules.
- Cross-check claims against land cover: if a caption mentions a river or lake but the cell has 0% water, delete that sentence. Do the same for forests, farmland and buildings.
- Drop the worst-matching image-caption pairs using a CLIP-style similarity score.

2d. Quality audit. Hand-check a random sample of 300 captions before and after cleaning. Report the share of captions with at least one false claim.

The final text for each cell = cleaned caption + fact sentences.

Inspired by: UrbanCLIP §3.1 and Appendix C (captioning with an LLM, a focused prompt, two-stage cleaning). The fact sentences and the land-cover cross-check are **our own addition**, designed to replace UrbanCLIP's manual student checking, which its authors themselves listed as future work.

7.3 Step 3: Stage A, image-text alignment (no labels)

The goal is a shared space where a cell's image embedding sits close to its own description (Concept 3.6).

1. **Image side:** the cell's 64 GSED numbers (frozen; we never change GSED).
2. **Text side:** a pretrained sentence-embedding model (frozen) turns each cell's text into a list of numbers.
3. **Trained part:** two small projection networks, one per side, map both into a shared 128-number space.
4. **Training signal:** a contrastive loss. In each batch, every image is pushed toward its own text and away from the other texts in the batch, and vice versa.
5. **Output:** for every cell (captioned or not), the projected image embedding. Captions are not needed at prediction time, because only the image side is used.

This is Option A1 from our earlier planning: cheap, and it runs on one GPU. We also keep **Option A0** (raw GSED, no Stage A) as a baseline to check whether Stage A helps.

Inspired by: UrbanCLIP §3.2–3.3 (image–text contrastive loss; using only the image encoder at prediction time). Using frozen GSED as the image encoder instead of a vision transformer is our simplification, borrowed from the embedding paper’s choice of precomputed embeddings.

7.4 Step 4: Stage B, night-light steering (proxy labels)

1. Train a small network on training-district cells to predict log nighttime light from the Stage A embedding. Layers: $128 \rightarrow 128 \rightarrow 64 \rightarrow 32 \rightarrow 1$.
2. Cut the network at the 32-number layer. This middle layer is the new **economy-aware embedding** for each cell.
3. Why it works: to predict lights, the network must keep features about buildings, roads and activity, and can ignore features about climate or vegetation (Concept 3.7).

We also build a version where Stage B takes raw GSED as input (no Stage A). This reproduces the embedding paper at our scale and lets us separate the effect of text from the effect of steering.

Inspired by: the embedding paper §2.2 (NTL mapper and 32-dimensional middle layer), which itself follows Jean et al. (2016). Feeding Stage A embeddings into Stage B, so both ideas are combined, is **our own addition**.

7.5 Step 5: Stage C, district GDP prediction (real labels)

1. Each cell’s 32-number embedding goes through a small per-cell network ($32 \rightarrow 32 \rightarrow 16$).
2. The 16-number outputs of all cells in a district are **added up** (sum-pooling, Concept 3.8).
3. A final network ($16 \rightarrow 16 \rightarrow 1$) turns the district total into predicted log district GDP.
4. Subtract log population to get predicted log GDP per capita.
5. Train only on training districts; use validation districts to decide when to stop.

Because there are only a few hundred districts, overfitting is a real risk. We use small layers, weight decay (a penalty on large weights), dropout (randomly switching off units during training) and early stopping. As a simpler check, we also fit a **ridge regression** (a linear model with a penalty) on the average of each district’s cell embeddings.

Inspired by: the embedding paper §2.3 (two-stage GDP mapper with sum-pooling across spatial units; the same layer sizes). Applying it to districts made of 1 km cells, instead of countries made of 10 km cells, is our adaptation. The frozen-encoder-plus-small-predictor idea is shared with UrbanCLIP §3.4.

7.6 Step 6: NTL-only benchmark and hybrid blend

1. **NTL-only model:** per district, compute log NTL per person, then fit a quadratic regression of log GDP per capita on it, using training districts only.
2. **Hybrid:** final estimate = $\lambda \times (\text{our model}) + (1 - \lambda) \times (\text{NTL-only model})$.
3. Try $\lambda = 0, 0.25, 0.5, 0.75$ and 1 on validation districts and keep the best.

Inspired by: the embedding paper §2.4 and §3.2 (convex combination with a λ grid), whose NTL benchmark follows Hu and Yao (2022).

7.7 Step 7: Maps and text explanations

1. **District map:** predicted versus official log GDP per capita, and the gap between them.
2. **Cell-level activity map:** the Stage B predicted-light value per 1 km cell, shown as a relative activity index. It is for display only, since there are no cell-level GDP labels to validate it.
3. **Text explanations:** for a chosen district, retrieve the training captions whose embeddings are most similar to its cells (cosine similarity in the Stage A shared space), and list words that are more common in high-prediction districts than in low ones (for example "industrial", "high-rise", "irrigated fields").

Inspired by: UrbanCLIP §4.4.2 (cross-city similarity search) and §4.5 (captions as human-readable explanations). The word-level comparison is **our own addition**; neither paper explains predictions with words.

8. Evaluation plan

Every model is judged on districts it has never seen, in any stage. This is the single most important rule of the project.

8.1 Spatial five-fold cross-validation

1. Group neighbouring districts into about 25 compact spatial blocks, so that folds are geographic chunks, not scattered districts (Concept 3.9).

2. Split the blocks into 5 folds, balanced so each fold has a similar mix of urban, peri-urban and rural districts.
3. In each round: 3 folds train, 1 fold validates, 1 fold tests. Rotate 5 times so every district is tested exactly once.
4. **Stages A, B and C are all retrained in every round, using training districts only.** Stage B uses nighttime lights, which are strongly linked to GDP, so letting it see test districts would leak a close stand-in for the test answers.
5. The validation fold is used for every choice (stopping time, λ , learning rate). The test fold is used only once per round, at the end.

Inspired by: the embedding paper §2.5 (five groups rotating as train, validation and test). Splitting by spatial blocks rather than randomly is our correction of UrbanCLIP's random 60/20/20 split.

8.2 Metrics

All metrics are computed on log GDP per capita of test districts, pooled across the 5 rounds.

- **R^2 , RMSE and MAE** (Concept 3.11), reported as mean and spread across rounds.
- **Validation regression:** regress official log GDP per capita on the predicted value. A slope near 1 and an intercept near 0 mean the estimates are not systematically too high or too low.

Inspired by: UrbanCLIP §4.1.3 (R^2 , RMSE, MAE) and the embedding paper §3.2 (validation regression and MSE).

8.3 Baselines and ablations

Each row is trained and tested with the same folds, so differences are fair.

Variant	What it removes or adds	Answers	Inspired by
NTL only (quadratic)	Satellite lights only	Benchmark to beat	Embedding paper; Hu and Yao (2022)
Land-cover percentages only	A simple, interpretable input	Is a simple model enough?	Our addition

Variant	What it removes or adds	Answers	Inspired by
Raw GSED (A0) → Stage C	No text, no steering	Base value of embeddings	Embedding paper
Raw GSED → Stage B → Stage C	Steering, no text	RQ1 (reproduces the embedding paper)	Embedding paper
Stage A → Stage C	Text, no steering	RQ2 without steering	UrbanCLIP
Stage A → Stage B → Stage C	Full model, no POI/roads	RQ2 with steering	Our combination
Full model + POI/roads	Everything	RQ3	UrbanCLIP §3.5
Full model, uncleaned captions	Skip Step 2c	Value of cleaning	UrbanCLIP §4.3.2
Full model, fact sentences only	No VLM captions	Is the VLM needed at all?	Our addition
Best model blended with NTL (λ grid)	Hybrid	RQ4	Embedding paper §2.4

8.4 Error analysis

The gap between our estimate and official GDP mixes our model's error with errors in the official data (Concept 3.10). We look at where gaps are large and what they relate to.

1. Regress the absolute error of each test district on log official GDP per capita and on its urban share.
2. If a data-quality indicator exists for districts (for example, whether figures are directly compiled or estimated), add it. If none exists, we say so and analyse income only.
3. Plot absolute error against income with a LOWESS curve (a smooth line following the local average), separately for our model and the NTL-only model.
4. List the 10 districts with the largest gaps and inspect their images and captions to suggest reasons (for example mining, ports, or large service sectors).

Inspired by: the embedding paper §2.6 and §3.4 (regression of absolute residuals, LOWESS plot against the SPI index) and §4 (careful interpretation of large gaps as "candidates", not proof).

8.5 Uncertainty

Train Stage C five times with different random starts. Report the mean prediction and the spread (minimum to maximum) per district. Check whether districts with a large spread also tend to have larger errors.

Inspired by: our own addition; neither paper reports uncertainty.

8.6 Optional extension: growth

If time allows, repeat the pipeline for a second year and test whether predicted changes match official changes, using district fixed effects (Concept 3.14). Based on the embedding paper, we expect NTL to beat embeddings here.

Inspired by: the embedding paper §3.2.2 and Table 4.

9. Requirements coverage

Of the 23 requirements in the full pipeline we designed (combining both papers), this term project fully meets 12, partly meets 6 and leaves out 5. The ones left out need heavy computing, more years of data, or more targets than one term allows.

"Full pipeline" means the complete design discussed during planning: data collection, text generation, extra map data, three learning stages, and a full evaluation. Status can be changed in the table as the project evolves.

ID	Requirement (full pipeline)	Status	How it is met, or why not	Where
R1	Collect satellite imagery or embeddings, and land-cover data	Fulfilled	GSED embeddings, high-resolution tiles for captions, ESA WorldCover	§6, §7.1
R2	Generate text descriptions with an LLM or small VLM	Fulfilled	Open VLM captions for about 10,000 cells, plus template fact sentences	§7.2
R3	Clean and refine the text	Fulfilled	Rule-based filtering, land-cover cross-check, similarity filtering, 300-caption audit	§7.2

ID	Requirement (full pipeline)	Status	How it is met, or why not	Where
R4	Add POI data	Fulfilled	OpenStreetMap POI counts, as features and in fact sentences	§6, §7.1
R5	Add road network data	Fulfilled	OpenStreetMap road length by type	§6, §7.1
R6	Align images and text with contrastive learning	Fulfilled	Option A1: frozen encoders, trained projections, contrastive loss	§7.3
R7	Fine-tune the image and text encoders (full UrbanCLIP model)	Not fulfilled	Needs large GPUs and weeks of tuning; replaced by frozen GSED plus small projections	§5.3
R8	Captioning (language-modelling) loss	Not fulfilled	Requires a text-generating decoder; out of scope for one term	§5.3
R9	Deep multimodal fusion (cross-attention)	Partially fulfilled	Replaced by simpler fusion: POI and roads enter as numbers and as text	§7.1, §7.2
R10	Steer embeddings toward economics with nighttime lights	Fulfilled	Stage B NTL mapper, 32-number middle layer	§7.4
R11	Predict the target from district-level labels	Fulfilled	Stage C sum-pooling from 1 km cells to districts	§7.5
R12	Blend with an NTL-only model	Fulfilled	Quadratic NTL model and λ grid chosen on validation	§7.6
R13	Train, validation and test split without leakage	Fulfilled	Five-fold spatial block cross-validation; all stages retrained per fold	§8.1
R14	Baselines and ablations	Fulfilled	Ten variants on identical folds	§8.3

ID	Requirement (full pipeline)	Status	How it is met, or why not	Where
R15	Error analysis by data quality and income	Partially fulfilled	Income analysis is done; a district-level data-quality indicator may not exist	§8.4
R16	Uncertainty estimates	Partially fulfilled	Simple five-run ensemble for Stage C only; no calibrated intervals	§8.5
R17	Growth (change over time) estimation	Partially fulfilled	Optional second year with fixed effects, only if time allows	§8.6
R18	Fine-resolution (1 km) maps	Partially fulfilled	Map is produced, but cannot be validated without cell-level labels	§7.7
R19	Transfer to unseen regions	Partially fulfilled	Spatial blocks test unseen areas within one region, not across regions or countries	§8.1
R20	Explain predictions with text	Fulfilled	Similar-caption retrieval and word comparison	§7.7
R21	Several targets (population, wealth, carbon)	Not fulfilled	One target keeps the project focused; others use the same pipeline later	§5.3
R22	Time-based split (train on past years, test on later years)	Not fulfilled	Single main year; needs multi-year data	§5.1
R23	Interactive web application	Not fulfilled	Not needed for the research questions	§5.3

10. Tools, compute and timeline

The project runs on free tools and a single free cloud GPU, over 12 weeks.

10.1 Tools

Task	Tool
Download GSED, VIIRS and WorldCover	Google Earth Engine Python API (with geemap)
Grid, districts, POI and roads	GeoPandas, OSMnx or pyrosm
Captions	Hugging Face transformers with an open VLM (e.g., Qwen2.5-VL 3B or Florence-2)
Text embeddings	sentence-transformers (a small pretrained model)
Stages A, B, C	PyTorch
Ridge, metrics, regressions, LOWESS	scikit-learn, statsmodels
Maps and figures	GeoPandas, matplotlib

10.2 Compute (approximate)

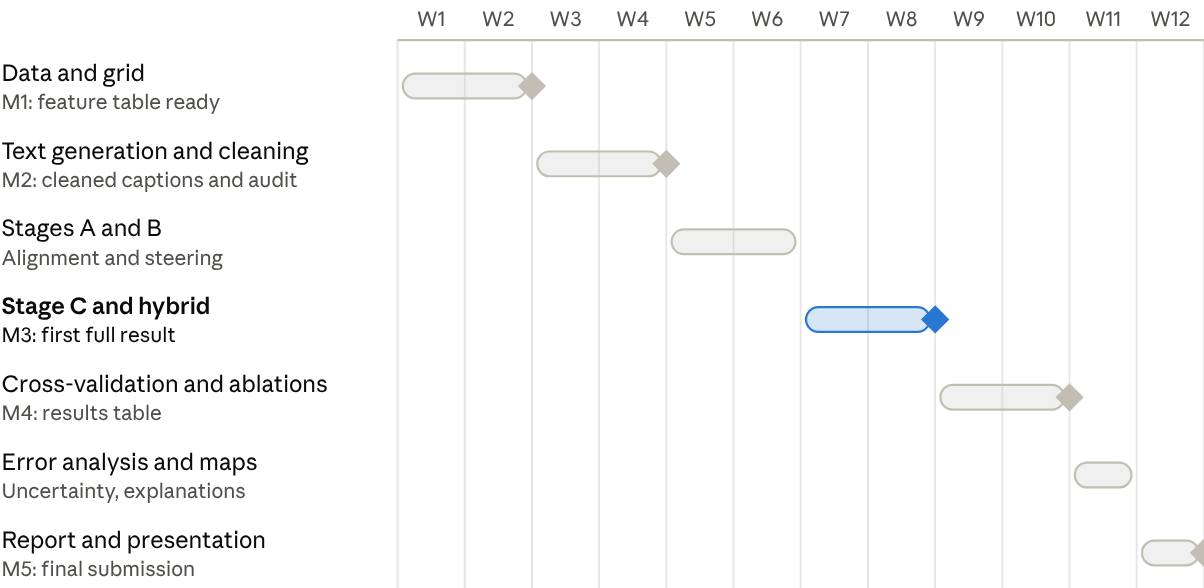
- **Captioning** is the heaviest step. A small VLM on a free T4 GPU (Google Colab or Kaggle) takes roughly 2–4 seconds per image, so 10,000 images need about 6–11 GPU hours, spread over several sessions.
- **Stages A–C** are small networks on tables of numbers. One full five-fold run should take minutes to an hour, even for all ten ablation variants.
- **Earth Engine exports** of 1 km averages may take hours per layer but run on Google's servers.

These are estimates; they will be checked in week 3 with a 200-image test run.

10.3 Timeline

First full result by week 8 leaves four weeks to test and write

Diamonds mark milestone gates; weeks counted from the project start



12-week plan · 7 phases, 5 gates

If M3 slips past week 8, the growth extension (§8.6) is dropped first, then the POI and roads ablation.

11. Risks and mitigations

The biggest risk is too few districts with reliable GDP; it is checked first, in week 1.

Risk	Effect	Mitigation
Too few districts with official GDP (under 150)	Stage C overfits; unstable results	Widen the study area; rely more on ridge regression; report spread across folds
High-resolution tiles unavailable or licence-restricted	No visual captions	Fall back to fact sentences only (already an ablation row), or caption Sentinel-2 true-colour images at lower quality

Risk	Effect	Mitigation
VLM captions are generic or often wrong	Text adds little	Stronger prompt, land-cover cross-check, report audit error rate; a negative RQ2 result is still valid
Captioning takes longer than planned	Delay in weeks 3–4	Reduce the sample to 5,000 cells; use a smaller VLM such as Florence-2
Official GDP has its own errors	Honest models look worse	Error analysis (§8.4) treats large gaps as candidates, not proof
Night-light saturation in big cities	Stage B learns little in the brightest areas	Keep raw-GSED variants; check errors by urban share
District boundaries in GDP data differ from boundary files	Cells assigned to the wrong district	Use boundaries from the same source as the GDP data; match names manually where needed
OpenStreetMap is incomplete in rural areas	POI features biased toward cities	Include a mapping-completeness check (e.g., road length vs built-up share); interpret RQ3 with care

12. Expected outcomes and deliverables

The project will show, with honest spatial testing, how much text and night-light steering each add to satellite-based GDP estimates, and whether they add more together.

12.1 Expected results

- Night-light steering improves raw embeddings (RQ1), as in the embedding paper.
- Text gives a small, measurable gain (RQ2); it may be larger in cities than in rural districts.
- The hybrid with NTL is the most accurate single estimate (RQ4).
- Errors grow with income, and are smaller for our model than for NTL alone (RQ5).

- Scores will be lower than UrbanCLIP's reported numbers, because spatial splitting removes leakage. This is expected and will be explained.

12.2 Deliverables

1. **Code repository** with a README to rerun every step.
2. **Cell feature table** and **cleaned caption set** with the 300-caption audit results.
3. **Results table** for all baselines and ablations, with mean and spread over five folds.
4. **Figures:** pipeline diagram, predicted vs official scatter plot, error vs income LOWESS plot, district gap map, 1 km activity map.
5. **Explanation examples:** three to five districts, each with similar captions and key words.
6. **Final report** (about 12–15 pages) and a presentation.

13. References

The two main source papers come first; the rest are works cited inside them that this proposal also relies on.

1. Yan, Y., Wen, H., Zhong, S., Chen, W., Chen, H., Wen, Q., Zimmermann, R. and Liang, Y. (2024). UrbanCLIP: Learning text-enhanced urban region profiling with contrastive language-image pretraining from the web. *Proceedings of the ACM Web Conference 2024 (WWW '24)*. [doi:10.1145/3589334.3645378](https://doi.org/10.1145/3589334.3645378)
2. Yue, X., Zhao, Z. and Hu, K. (2026). Estimating economic activity from satellite embeddings. *Applied Sciences*, 16, 582. [doi:10.3390/app16020582](https://doi.org/10.3390/app16020582)
3. Brown, C. F. et al. (2025). AlphaEarth Foundations: An embedding field model for accurate and efficient global mapping from sparse label data. arXiv:2507.22291.
4. Google. Satellite Embedding V1 (annual). [Earth Engine data catalog](https://earthengine.google.com/data-catalog/)
5. Henderson, J. V., Storeygard, A. and Weil, D. N. (2012). Measuring economic growth from outer space. *American Economic Review*, 102, 994–1028.
6. Jean, N., Burke, M., Xie, M., Davis, W. M., Lobell, D. B. and Ermon, S. (2016). Combining satellite imagery and machine learning to predict poverty. *Science*, 353, 790–794.
7. Hu, Y. and Yao, J. (2022). Illuminating economic growth. *Journal of Econometrics*, 228, 359–378.
8. Radford, A. et al. (2021). Learning transferable visual models from natural language supervision (CLIP). *ICML 2021*.

9. Yu, J. et al. (2022). CoCa: Contrastive captioners are image-text foundation models. *Transactions on Machine Learning Research*.
10. Gao, P. et al. (2023). LLaMA-Adapter V2: Parameter-efficient visual instruction model. arXiv:2304.15010.
11. Xi, Y., Li, T., Wang, H., Li, Y., Tarkoma, S. and Hui, P. (2022). Beyond the first law of geography: Learning representations of satellite imagery by leveraging point-of-interests (PG-SimCLR). *Proceedings of the ACM Web Conference 2022*.
12. Miller, H. J. (2004). Tobler's first law and spatial analysis. *Annals of the Association of American Geographers*, 94, 284–289.
13. Elvidge, C. D., Zhizhin, M., Ghosh, T., Hsu, F. C. and Taneja, J. (2021). Annual time series of global VIIRS nighttime lights derived from monthly averages: 2012 to 2019. *Remote Sensing*, 13, 922.
14. Dang, H. A. H., Pullinger, J., Serajuddin, U. and Stacy, B. (2023). Statistical performance indicators and index: A new tool to measure country statistical capacity. *Scientific Data*, 10, 146.